

Joint Models for Longitudinal and Survival Analysis Data from the Bayesian approach

Final project assignment

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Abstract

Survival analysis and longitudinal data are truly widespread types of data when working with clinical trials. Nonetheless, they are normally treated separately. So, what happens when we have time to event and longitudinal data in the same dataset? Joint Models, this is the answer. Joint Models [10] are tools that allow us to deal with both types of data at the same time. In this work, a rigorous theoretical explanation about joint models will be provided. The Bayesian model used when working with these objects will be exposed in detail. In the second part of the work, theoretical knowledge will be put into practice through the analysis of the AIDS data set. The software used is the *JMbayes2* R package [12]. This work will demonstrate that joint models are useful when comparing the effect of two treatments, taking into account fixed covariates as well as longitudinal outcomes.

Resum

L'Anàlisi de la Supervivència i l'Anàlisi de dades Longitudinals són camps molt extesos quan es treballa amb assajos clínics. Nogensmenys, aquests tipus de dades són normalment tractades per separat. Així doncs, què passa quan tenim dades de supervivència i longitudinals en la mateixa base de dades? *Joint Models*, és la resposta. Els *Joint Models* [10] són eines que ens permeten tractar amb els dos tipus de dades alhora. En aquest treball es fa una explicació teòrica rigurosa sobre els *Joint Models*. El model bayesià que es fa servir quan es treballa amb aquests objectes s'exposarà en detall. A la segona part del treball, el coneixement teòric es posarà en pràctica a través de l'anàlisi de la base de dades AIDS. El *software* emprat és el paquet d'R *JMbayes2* [12]. En aquest treball es demostra que els *joint models* són útils per comparar l'efecte de dos tractaments, tenint en compte alhora covariables fixes i longitudinals.

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1 Introduction

Time to event and longitudinal data are widely used in clinical trials and other health-based studies. Our goal in this work is to mix these two frameworks into the so-called *Joint Models*, and analyze data in which we have time to event data as well as longitudinal measurements. The main motivation of this work is theoretical. The aim is to study the mathematics behind *Joint Models*, understanding how Survival Analysis and Longitudinal data analysis join forces to create a theoretical framework that allows us to work with data that, in the world of clinical data, appears very frequently. Furthermore, the whole theoretical knowledge will be applied in a case study, in which the Bayesian capabilities will be exploited to get the best model possible.

On the one hand, Survival Analysis is an important statistical field studying the time until an event occurs. Plenty of applications exist for this methodology, one of the most widespread usage is in the clinical trials context. Although Survival Analysis had been initially developed under the frequentist approach, nowadays one can find in the literature many works on Survival Analysis using the Bayesian paradigm, see [4] or [6]. On the other hand, longitudinal data appears when taking several measurements during a period time of an individual. Clearly, the correlation among these observations needs to be treated properly. Longitudinal data analysis proposes a mixed-effects model that accounts for the correlation in this type of data. Moreover, a random effects model is more intuitive and understandable when working under the Bayesian approach. Of course, longitudinal data appears constantly in medical studies.

But, what happens if we have data such that besides time until an event of clinical interest, several biomarkers are measured during a period of time for each individual? Notice that this is not an uncommon case, or a very specific type of data, but a data structure that appears usually when dealing with clinical data. Another theoretical framework is needed to cope with this kind of data. *Joint Models* [10] is the answer.

In the first part of the work we will expose the main features about Survival Analysis and the main differences when dealing with this field under the Bayesian approach with respect the more classical frequentist approach. Then, the longitudinal data framework will be presented. To conclude the most theoretical part the Joint Models will be presented in a mathematical and rigorous manner. The main assumptions taken will be discussed and the Bayesian model will be stated. Then, we will jump to the practical part. We will analyze the *AIDS* data set. Data from a randomized clinical trial in which both longitudinal and survival data were collected to compare the efficacy and safety of two antiretroviral drugs in treating patients who had failed or were intolerant of zidovudine (AZT) therapy. By means of the *JMbayes2* R package [12], several Joint Models will be fitted to the data.

2 Survival Analysis: main features

In this initial section the main features on Survival Analysis under the Bayesian approach will be exposed. For further information on the subject we recommend [4] or [6], the latter being a more recent work.

Let T be the time until an event of interest, say $\mathcal{E}$, occurs. Let C be the random variable accounting for the censored times. Then, we observe $T_i^* = \min\{T_i, C_i\}$ for each patient $i = 1, \dots, n$. As well as, we define $\delta_i = \mathbf{1}\{T_i \leq C_i\}$ as the censoring indicator. Having defined these quantities, the next step is to define some basic functions in the context of Survival Analysis. We denote by $F(t) = P(T \leq t)$, the cumulative distribution function of the random variable T . We define the survival function as follows:

$$S(t) = 1 - F(t) = P(T^* > t). \quad (1)$$

Besides the survival function, the hazard function is a key function when dealing with time to event data. The hazard function is defined as the limit

$$h(t) = \lim_{dt \rightarrow 0} \frac{P(t \leq T^* < t + dt | T^* \geq t)}{dt}, \quad (2)$$

and is interpreted as the instantaneous risk of suffering the event $\mathcal{E}$. There is a useful relation between the two functions defined above, which is the following

$$S(t) = \exp\left(-\int_0^t h(s)ds\right). \quad (3)$$

2.1 Likelihood for right-censored data

In order to work with a Bayesian model it is essential to present the likelihood of the model. Notice that until now we do not have assumed any parametric model for our random variables T and C . A parametric model will not be assumed in any moment, as the reader will see in the following sections it is not needed to deal with our data. In particular, we do not need to assume a parametric model to state the likelihood of a survival model under right-censored data. Nonetheless, another hypothesis should be taken, it is the non-informative censoring hypothesis. This is a key assumption in survival analysis problems, and it states that time to event T and the censoring times C are unrelated.

So, assuming non-informative censoring as well as that the cumulative distribution functions of T and C are functionally independent, the likelihood function of a right-censored survival model is

$$L = \prod_{i=1}^n h(t_i^*)^{\delta_i} S(t_i^*). \quad (4)$$

Then, while the contribution of an individual with failure time observed is $h(t_i^*)S(t_i^*)$, the contribution of an individual with failure non-observed is just the survival $S(t_i^*)$.

2.2 The Cox model

The Cox model is a relative risk regression model. It is the most widespread model in Survival Analysis. Proportionality of hazard functions is assumed. The Cox model expression is

$$h_i(t|\mathbf{w}_i) = h_0(t) \exp(\boldsymbol{\gamma}^\top \mathbf{w}_i), \quad (5)$$

where $\mathbf{w}_i = (w_{i1}, \dots, w_{ip})$ denotes the vector of covariates that are assumed to be associated with the hazard of each subject, $\boldsymbol{\gamma}$ denotes the vector of regression coefficients and $h_0(t)$ the baseline hazard function.

When working under the frequentist approach, the estimation of the regression coefficients is done by means of the so-called *partial likelihood* [3]. Moreover, the baseline hazard function $h_0(t)$ remains unspecified. However, from the Bayesian point of view does not make sense to work with a partial likelihood, it is, without the complete likelihood. So, when working under the Bayesian approach the baseline hazard function should be specified, for instance by assuming a parametric model for T . Another option, which allows more flexibility to model the baseline hazard, is to use a piece-wise hazard function, or to use Splines. Read more about Cox model under the Bayesian approach on [1] or in the first chapter of [6].

3 Longitudinal Data Analysis: the linear mixed model

Imagine the next scenario: we have collected different biomarkers for each patient during a period of time. Clearly, there is a correlation between the observations of the same individual that we have to take into account when modeling the data. Longitudinal Data Analysis emerges to deal with this type of data. Furthermore, random effects are used in the Longitudinal Analysis field, which makes the usage of the Bayesian approach more intuitive than the frequentist one. This is because under the Bayesian point of view all the parameters involved in the model are random variables. Contrarily, in the frequentist approach parameters are treated as deterministic quantities. In addition to parameters, random effects are random variables. So, working under the Bayesian approach seems to be more natural.

In this work we focus on normally distributed longitudinal outcomes, we use a *linear mixed-effects* model to describe the subject-specific longitudinal trajectories. For the longitudinal process, we let $\mathbf{y}_i = (y_{1i}, \dots, y_{n_i i})$ denote the $n_i \times 1$ longitudinal response vector for the i -th subject, with element y_{ij} denoting the value of the longitudinal outcome taken at time point t_{ij} , $j = 1, \dots, n_i$. We have the following model

$$\begin{aligned} y_i(t) &= m_i(t) + \varepsilon_i(t) = \mathbf{x}_i^\top(t) \boldsymbol{\beta} + \mathbf{z}_i^\top(t) \mathbf{b}_i + \varepsilon_i(t) \\ \mathbf{b}_i &\sim \mathcal{N}(\mathbf{0}, \mathbf{D}), \quad \varepsilon_i(t) \sim \mathcal{N}(0, \sigma^2), \end{aligned}$$

where $y_i(t)$ denotes the value of the longitudinal outcome at any particular time point t , $\mathbf{x}_i(t)$ and $\mathbf{z}_i(t)$ denote the time-dependent design vectors for the fixed-effects $\boldsymbol{\beta}$ and for

the random effects b_i , respectively, and $\varepsilon_i(t)$ the corresponding error terms that are assumed independent of the random effects, and $\text{Cov}(\varepsilon_i(t), \varepsilon_i(t')) = 0$ for $t' \neq t$.

4 Joint models

In this section we have followed mainly [10]. Despite being a frequentist reference, the reader can find there the whole theoretical framework of Joint Models clearly exposed, as well as several examples on R. Moreover, [11] has been used to present the Joint Models in the Bayesian context.

4.1 Model expression and main characteristics

Assume that we have data combining time-to-event and longitudinal data. Taking into account the survival and the longitudinal model introduced above, as well as using the previous notation, we can write the model expression of the most basic Joint model,

$$h_i(t | \mathcal{M}_i(t), w_i) = h_0(t) \exp \left\{ \gamma^\top w_i + \alpha m_i(t) \right\}, \quad t > 0, \quad (6)$$

where $\mathcal{M}_i(t) = \{m_i(s), 0 \leq s < t\}$ denotes the history of the true unobserved longitudinal process up to t . The parameter α quantifies the association between the true value of the marker at t and the hazard for an event at the same time point. Regarding the rest of components in (6), they were introduced in the previous sections and its interpretations are as explained above. Regarding the interpretation of the association parameter α , it is analogous to the interpretation of γ , it is a relative risk. In particular, $\exp(\gamma_j)$ denotes the ratio of hazards for one unit change in w_{ij} at any time t , whereas $\exp(\alpha)$ denotes the relative increase in the risk for an event at time t that results from one unit increase in $m_i(t)$ at the same time point.

Note that in the relative risk model (6) the risk for an event at time t depends only on the current value of the time-dependent marker $m_i(t)$. Nonetheless, using the relation between the survival and the hazard function (3) stated before, one can see easily that this characteristic does not hold for the survival function,

$$S_i(t | \mathcal{M}_i(t), w_i) = \exp \left(- \int_0^t h_0(s) \exp \left\{ \gamma^\top w_i + \alpha m_i(s) \right\} ds \right), \quad (7)$$

which implies that the corresponding survival function depends on the whole covariate history $\mathcal{M}_i(t)$. This feature is important because the survival function appears in the likelihood of our model, as we will see in the following section.

To complete the specification of our Joint Model we need to discuss the choice for the baseline hazard function $h_0(\cdot)$. As we commented before, the baseline hazard usually is left unspecified in Survival Analysis. However, in the joint modeling context we need to specify it. It is possible to assume a parametric model, normally the Weibull, the log-normal or the Gamma are the distributions used. However, we can opt for a parametric, but flexible specification of the baseline hazard function. Use a piece-wise constant function is a good

option. Nonetheless, in this work a regression splines model approach will be used. The $\log(h_0(t))$ is expanded into B-splines basis functions for splines as follows:

$$\log h_0(t) = \gamma_{h_0,0} + \sum_{q=1}^Q \gamma_{h_0,q} B_q(t, v), \quad (8)$$

where $B_q(t, v)$ denotes the q -th basis function of a B-spline with knots $v_1, \dots, v_Q$ and γ_{h_0} the vector of spline coefficients. Increasing the number of knots we increase the flexibility in approximating $h_0(\cdot)$. Of course, a balance between bias and variance should be taken, as well as overfitting should be avoided when selecting the number of knots and the flexibility in our estimation. A standard rule of thumb is to keep the total number of parameters, including the parameters in the linear predictor in (6) and in the model for $h_0(\cdot)$, between 1/10 and 1/20 of the total number of events in the sample.

4.2 Likelihood of the model

In this work we use the so-called *Joint Likelihood formulation* (see fourth chapter of [10]). Despite two stage approaches can be used, since our goal is to build a Bayesian model, we work with a joint likelihood formulation. In addition, this is the main method used in the joint modeling literature.

There are three important assumptions to be made. Firstly, we assume conditional independence on the random effects $\mathbf{b}_i$. Secondly, we assume that the longitudinal responses of each subject are independent. Finally, we assume that given the observed history of longitudinal responses up to time s , the censoring mechanism and the visiting process are independent of the true event times and future longitudinal measurements at $t > s$. Taking these assumptions into account we can write

$$\begin{aligned} p(\mathbf{y}_i, T_i, \delta_i \mid \mathbf{b}_i, \boldsymbol{\theta}) &= p(\mathbf{y}_i \mid \mathbf{b}_i, \boldsymbol{\theta}) p(T_i, \delta_i \mid \mathbf{b}_i, \boldsymbol{\theta}) \\ p(\mathbf{y}_i \mid \mathbf{b}_i, \boldsymbol{\theta}) &= \prod_l p(y_{il} \mid \mathbf{b}_i, \boldsymbol{\theta}), \end{aligned} \quad (9)$$

where $\boldsymbol{\theta}^\top = (\boldsymbol{\theta}_t^\top, \boldsymbol{\theta}_y^\top, \boldsymbol{\theta}_b^\top)$ denotes the full parameter vector, with $\boldsymbol{\theta}_t$ denoting the parameters for the event time outcome, $\boldsymbol{\theta}_y$ the parameters for the longitudinal outcomes, and $\boldsymbol{\theta}_b$ the unique parameters of the random-effects covariance matrix. Notice that the first expression is due to the fact that we have assumed conditional independence on $\mathbf{b}_i$. It is, the random effects account for both the association between the longitudinal and event outcomes, and the correlation between the repeated measurements. On the second expression, we basically have the independence between the different longitudinal outcomes, assumed as well.

Under these assumptions, the likelihood contribution for the i -th subject conditional on the parameters and random effects takes the form:

$$\begin{aligned}
p(\mathbf{y}_i, T_i, \delta_i \mid \boldsymbol{\theta}, \mathbf{b}_i) &= \prod_{l=1}^{n_i} p(y_{il} \mid \mathbf{b}_i; \boldsymbol{\theta}_y) p(T_i, \delta_i \mid \mathbf{b}_i; \boldsymbol{\theta}_t, \boldsymbol{\beta}) p(\mathbf{b}_i; \boldsymbol{\theta}_b) \\
&\propto \left[(\sigma^2)^{-n_i/2} \exp \left\{ - \sum_l \left(y_{il} - \mathbf{x}_{il}^\top \boldsymbol{\beta} - \mathbf{z}_{il}^\top \mathbf{b}_i \right)^2 / 2\sigma^2 \right\} \right. \\
&\quad \times \left[\exp \left\{ \sum_q \gamma_{h_0,q} B_q(T_i, \mathbf{v}) + \boldsymbol{\gamma}^\top \mathbf{w}_i + \alpha m_i(T_i) \right\} \right]^{\delta_i} \\
&\quad \times \exp \left[- \exp(\boldsymbol{\gamma}^\top \mathbf{w}_i) \int_0^{T_i} \exp \left\{ \sum_q \gamma_{h_0,q} B_q(s, \mathbf{v}) + \alpha m_i(s) \right\} ds \right] \\
&\quad \times \det(\mathbf{D})^{-1/2} \exp \left(-\mathbf{b}_i^\top \mathbf{D}^{-1} \mathbf{b}_i / 2 \right) \Big].
\end{aligned} \tag{10}$$

Despite being a large formula, the reader can easily identify the different components of our model. After the proportionality symbol we have in the first line the density of the longitudinal outcomes. As exposed above, we work with normally distributed longitudinal outcomes. Then, we obtain the density of the different longitudinal outcomes which takes the shape of a normal distribution. In the second and third line we have the survival sub-model contribution. Observe that here we have used the relation between the survival and the hazard (7) and the likelihood function for right censored data (4). Finally, we have the density corresponding to the random effects, which is a multivariate normal, as we have assumed in the longitudinal sub-model. Hence, as the reader can observe, after assuming those hypothesis, the likelihood is simply the product of the contribution of the different components of the model for each individual: the longitudinal sub-model, the survival sub-model and the random effects.

4.3 Priors of the Bayesian model

Of course, a mandatory step when building a Bayesian model is choosing the priors for all the parameters involved in the model. We choose for all the parameters non-informative prior distributions. Specifically, we select:

- For the vector of fixed effects of the longitudinal sub-model $\boldsymbol{\beta}$: independent univariate diffuse normal prior.
- For the regression parameters of the survival model $\boldsymbol{\gamma}$: independent univariate diffuse normal prior.
- For the vector of spline coefficients for the baseline hazard γ_{h_0} : independent univariate diffuse normal prior.
- For the association parameter α : independent univariate diffuse normal prior.
- For the variance of the error terms σ^2 : inverse-Gamma prior.
- For covariance matrices: an inverse Wishart prior (the conjugate prior for the covariance matrix of a multivariate normal distribution.).

4.4 Functional forms

There are several possible formulations for the association structure between the longitudinal outcome and the risk for an event. In (6) we have the most simple functional form possible, it is, put directly $m_i(t)$ in the Cox model expression and the association parameter α . The general formulation, taking into account any functional form is

$$h_i(t) = h_0(t) \exp \left(\gamma^\top \mathbf{w}_i + \alpha^\top f(t, \mathbf{b}_i, \mathcal{M}_i(t)) \right), \quad (11)$$

where $f(\cdot)$ is a time-dependent function that may depend on the random effects and on the true longitudinal history, and α is a vector of association parameters. The most widely used functional forms are:

- *Time-dependent slopes*: joint model in which the risk depends on both the current true value of the trajectory and the slope of the true trajectory at time t . The relative risk survival sub-model takes the form,

$$h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha_1 m_i(t) + \alpha_2 m'_i(t) \right\}, \quad t > 0, \quad (12)$$

where $m'_i(t) = d(\mathbf{x}_i^\top(t)\boldsymbol{\beta} + \mathbf{z}_i^\top(t)\mathbf{b}_i)/dt$. The interpretation of parameter α_1 is as before. Parameter α_2 measures the association between the value of the slope of the true longitudinal trajectory at time t and the risk for an event at the same time point, provided that $m_i(t)$ remains constant.

- *Cumulative effects*: we will account for the cumulative effect of the longitudinal outcome by including in the linear predictor of the relative risk sub-model the integral of the longitudinal trajectory from baseline up to time t . The survival sub-model is as follows

$$h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha \int_0^t m_i(s) ds \right\}, \quad t > 0, \quad (13)$$

where for any particular time point t , α measures the strength of the association between the risk for an event at time point t and the area under the longitudinal trajectory up to the same time t .

- *Shared random effects*: one of the standard formulations of joint models includes in the linear predictor of the risk model only the random effects of the longitudinal sub-model, obtaining,

$$h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha \mathbf{b}_i \right\}, \quad t > 0, \quad (14)$$

where α denotes a vector of association parameters each one measuring the association between the corresponding random effect and the hazard for an event.

4.5 Dynamic Individualized Predictions

The so-called Dynamic Individualized Predictions are individualized predictions for both survival and longitudinal outcomes, that have dynamical nature. Under the Bayesian model

stated above, we can derive these quantities which are interesting from the personalized medicine point of view. See [9] and [8] for further information on dynamic individualized predictions.

Assume that we have the data $\mathcal{D}_n = \{T_i, \delta_i, \mathbf{y}_i; i = 1, \dots, n\}$, we are interested in deriving predictions for a new subject j from the same population that has provided a set of longitudinal measurements $\mathcal{Y}_j(t) = \{y_j(s); 0 \leq s \leq t\}$, and has a vector of baseline covariates $\mathbf{w}_j$. Since biomarker measurements have been recorded up to t , survival of this subject is guaranteed up to t . Then, it is more meaningful to focus on the conditional subject-specific predictions, given survival up to t . In particular, for any time $u > t$ we are interested in the probability that this new subject j will survive at least up to u , it is,

$$\pi_j(u | t) = P\left(T_j^* \geq u \mid T_j^* > t, \mathcal{Y}_j(t), \mathbf{w}_j, \mathcal{D}_n\right)$$

Similarly, for the longitudinal outcome we are interested in the predicted longitudinal response at u , so we have

$$\omega_j(u | t) = E\left\{y_j(u) \mid T_j^* > t, \mathcal{Y}_j(t), \mathcal{D}_n\right\}.$$

The reader can observe that $\pi_j(u | t)$ and $\omega_j(u | t)$ have a dynamic nature, since when new information is recorded for patient j at time $t' > t$, we can update these predictions to obtain $\pi_j(u | t')$ and $\omega_j(u | t')$, and therefore proceed in a time dynamic manner.

Using the Bayesian framework stated in the previous sections, the estimation of these quantities is based on the corresponding posterior predictive distribution. For the survival outcome we have

$$\begin{aligned} \pi_j(u | t) &= \int P\left(T_j^* \geq u \mid T_j^* > t, \mathcal{Y}_j(t), \boldsymbol{\theta}\right) p(\boldsymbol{\theta} \mid \mathcal{D}_n) d\boldsymbol{\theta} \\ &= \int \left(\int P\left(T_j^* \geq u \mid T_j^* > t, \mathbf{b}_j, \boldsymbol{\theta}\right) p\left(\mathbf{b}_j \mid T_j^* > t, \mathcal{Y}_j(t), \boldsymbol{\theta}\right) d\mathbf{b}_j \right) p(\boldsymbol{\theta} \mid \mathcal{D}_n) d\boldsymbol{\theta} \\ &= \int \int \frac{S_j\{u \mid \mathcal{M}_j(u, \mathbf{b}_j), \boldsymbol{\theta}\}}{S_j\{t \mid \mathcal{M}_j(t, \mathbf{b}_j), \boldsymbol{\theta}\}} p\left(\mathbf{b}_j \mid T_j^* > t, \mathcal{Y}_j(t), \boldsymbol{\theta}\right) p(\boldsymbol{\theta} \mid \mathcal{D}_n) d\boldsymbol{\theta} d\mathbf{b}_j. \end{aligned}$$

In addition, for the longitudinal outcome we have

$$\begin{aligned} \omega_j(u | t) &= \int E\left\{y_j(u) \mid T_j^* > t, \mathcal{Y}_j(t), \boldsymbol{\theta}\right\} p(\boldsymbol{\theta} \mid \mathcal{D}_n) d\boldsymbol{\theta} \\ &= \int \int \left(\mathbf{x}_j^\top(t) \boldsymbol{\beta} + \mathbf{z}_i^\top(t) \mathbf{b}_j\right) p\left(\mathbf{b}_j \mid T_j^* > t, \mathcal{Y}_j(t), \boldsymbol{\theta}\right) p(\boldsymbol{\theta} \mid \mathcal{D}_n) d\boldsymbol{\theta} d\mathbf{b}_j. \end{aligned}$$

In order to get the previous expressions we have taken profit of the conditional independence assumption explained before. By means of the MCMC sample from the posterior distribution and the equations stated above, we can obtain simulations of $\pi_j(u | t')$ and $\omega_j(u | t')$. To approximate $\pi_j(u | t')$ by simulation, first of all, we must set the number of simulations, M , then for $l = 1, \dots, M$ we do:

1. Simulate $\theta^{(l)}$ from $[\theta|\mathcal{D}_n]$ (it is, the posterior distribution of θ).
2. Simulate $b_j^{(l)}$ from $[b_j|T_j^* > t, \mathcal{Y}_j(t), \theta^{(l)}]$.
3. Compute $\pi_j^{(l)}(u | t') = S_j \left\{ u | \mathcal{M}_j \left(u, b_j^{(l)} \right), \theta^{(l)} \right\} / S_j \left\{ t' | \mathcal{M}_j \left(t', b_j^{(l)} \right), \theta^{(l)} \right\}$.

With this, we can use the M Monte Carlo samples and compute the estimate

$$\hat{\pi}_j(u | t') = \text{mean} \left\{ \pi_j^{(l)}(u | t'), l = 1, \dots, M \right\},$$

credibility intervals can be computed as well.

Similarly, in order to compute $\omega_j(u | t')$ by simulation we follow the scheme:

1. Simulate $\theta^{(l)}$ from $[\theta|\mathcal{D}_n]$ (it is, the posterior distribution of θ).
2. Simulate $b_j^{(l)}$ from $[b_j|T_j^* > t, \mathcal{Y}_j(t), \theta^{(l)}]$.
3. Compute $\omega_j^{(l)}(u | t') = \mathbf{x}_j^\top(t')\beta^{(l)} + \mathbf{z}_i^\top(t')b_j^{(l)}$.

for $l = 1, \dots, M$. Proceeding exactly as before we can compute a point estimate by means of the mean, as well as credibility intervals. Fortunately, these computations are done by the software we will be using in the case study part. Nonetheless, it is important to explain the theoretical procedure behind this type of predictions. It is important to comment that the simulation scheme to deal with the dynamic individualized predictions is has been taken from a set of slides ¹ by Dimitris Rizopoulos.

5 Bayesian Model Averaging

Predictions are traditionally based in one model, this model is chosen after applying a selection model criterion. Nonetheless, by using just one model we are ignoring the possibility that more than one model can explain the uncertainty behind our problem. It is, there may be more than one true model. Thanks to the flexibility of the Bayesian theoretical framework we can exploit the predictive power of several models by means of the so-called *Bayesian model averaging* technique. See, for instance, chapter 10 of [7] for further information on the topic.

However, in this section we add a brief theoretical discussion on the following question: can we use Bayesian model averaging to do inference? Let us introduce the framework in which we work. Assume that we are under the joint modeling framework, and assume that all the covariates to be added in the model (either in the longitudinal sub-model or the survival sub-model) are fixed. Then, one can state the following K joint models:

¹See the set of slides on <https://www.drizopoulos.com/courses/EMC/ESP72.pdf>

$$\begin{aligned}
 M1: \quad & h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha m_i(t) \right\}, \\
 M2: \quad & h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha_1 m_i(t) + \alpha_2 m'_i(t) \right\}, \\
 M3: \quad & h_i(t | \mathcal{M}_i(t), \mathbf{w}_i) = h_0(t) \exp \left\{ \gamma^\top \mathbf{w}_i + \alpha \int_0^t m_i(s) ds \right\}, \\
 & \dots \\
 M_K: \quad & h_i(t) = h_0(t) \exp \left(\gamma^\top \mathbf{w}_i + \alpha^\top f(t, \mathbf{b}_i, \mathcal{M}_i(t)) \right).
 \end{aligned}$$

Where $p(M_j) = 1/K$ for all $j = 1, \dots, K$. Then, we are in the very particular case in which we have the same covariates in all the models, with the exception of the association parameter α , that takes different shapes, and we only change the functional form used in each model. Under this scenario, the Bayesian model averaging could be used for prediction, as is normally done. Nevertheless, in this specific case may also make sense to use model averaging for inference. Now, we introduce the different quantities needed to use the model averaging technique. Let $\mathcal{D}_n = \{T_i, \delta_i, \mathbf{y}_i; i = 1, \dots, n\}$ be the data used to fit the models $M_1, M_2, \dots, M_K$. The main metrics when working with Bayesian model averaging are the model weights, or the posterior probability for each model,

$$p(M_j | \mathcal{D}_n) = \frac{p(\mathcal{D}_n | M_j) p(M_j)}{\sum_{k=1}^K p(\mathcal{D}_n | M_k) p(M_k)}, \quad l = 1, \dots, K. \quad (15)$$

Moreover, $p(\mathcal{D}_n | M_j)$ is computed as follows

$$p(\mathcal{D}_n | M_j) = \int p(\mathcal{D}_n | \boldsymbol{\theta}_j) p(\boldsymbol{\theta}_j | M_j) d\boldsymbol{\theta}_j. \quad (16)$$

Once the integral above is solved for all the K models, the computation of the weights is straightforward. Nonetheless, the computation of this integral in our problem is unfeasible by means of analytic methods. There are several possibilities to compute the marginal likelihood of the model (16). The main difficulty in practice when using simulation is the fact that our priors are non-informative. Thus, a simulation of the non-informative priors could be difficult and derive to non-correct results. There is literature facing with this problem, as well as how to proceed to estimate (16) by simulation, see [2]. Assume that all the model weights have been computed, we purpose using these weights to obtain posterior distributions of the parameters in the model. Let γ_1 be the parameter of the survival sub-model associated with the treatment covariate. It is, we want to analyze whether a treatment have a significant effect with respect placebo (baseline treatment). With this goal in mind, we fit the K models above adding in all of these models the γ_1 parameter, and we compute the hazard ratio (HR) doing $\exp(\gamma_1)$. With this, we have the posterior distribution of the HR for each model. Let $\exp(\gamma_1)_1, \exp(\gamma_1)_2, \dots, \exp(\gamma_1)_K$ be the posterior distributions of the hazard ratio of γ_1 for each model. Now, using the model weights we can compute

$$HR_T = \sum_{j=1}^K \exp(\gamma_1)_j p(M_j | \mathcal{D}_n), \quad (17)$$

where HR_T is denoting the total hazard ratio posterior distribution, or the hazard ratio posterior distribution taking into account the contribution of all the models. Notice that HR_T is necessarily positive, since we are applying a convex combination ($\sum_{j=1}^K p(M_j|\mathcal{D}_n) = 1$, and $0 \leq p(M_j|\mathcal{D}_n) \leq 1$) to positive values. Thus, HR_T does not present any drawback to be the posterior distribution of $HR = \exp(\gamma_1)$. Moreover, the fact that we are using K models should reinforce the conclusions we draw. It is well-known that no model is completely correct, then, using K models to do the inference seems a good idea to produce robust results. In conclusion, if we obtain HR_T such that it is significantly greater than 1, say that the 95% credibility interval does not include 1 and is above 1. We can state, supported by K models and their respective weights, that the instantaneous risk of suffer an event is greater for those patients receiving the treatment with respect those patients receiving placebo, so the treatment is not working.

6 Missing data in Joint Models: main features

It is mandatory to add a discussion about missing data in the Joint modeling context. When dealing with data with longitudinal measurements, often occurs that there is missing data. It is not unusual for people to miss scheduled medical appointments. So, when working under this framework we have a specific type of missing data that should be managed. The study in depth of missing data, as well as missing data in longitudinal data, is out of the scope of this work. Nonetheless a brief discussion will be added in this section. See the second chapter of [10] or [5] for further information in this field.

Let us reduce to the dropout scenario, the subject fails one visit and then does not return anymore. This is a suitable scenario in our case, remember that we are working jointly with time to event data and longitudinal data. Obviously, when the event for the i -th subject is observed at T_i^* (either censored or observed), then the i -th subject no longer attends to the visits. Then, using the missing data notation in [10], we can denote T_i^* the time to dropout due to an event, and y_i^o, y_i^m the longitudinal measurement before T_i^* (observed) and the longitudinal measurement after T_i^* (missing), respectively.

Now, under the joint modeling framework we can distinguish between two types of dropout.

- If the subject drops out at T_i and $\delta_i = 0$, then we are under a missing completely at random (MCAR) or missing at random (MAR) dropout scenario.
- If the subject drops out at T_i and $\delta_i = 1$, then we are in the missing not at random (MNAR) dropout case.

In our case study we will be working under the MAR dropout scenario. We would like to comment that the missing data framework exposed in the references used in this part of the work are of a frequentist nature. Then, as a future research may be interesting to explore a Bayesian theoretical framework to deal with missing data under the Joint modeling framework, as well as to explore Bayesian missing data mechanisms to work with longitudinal data.

7 Software

When working with Survival Analysis in general, using Stan or other Bayesian software is not that easy, due to the fact that we must deal with censoring times. The reader can see an example of a Survival Analysis problem, specifically a Frailty model, done by myself under the Bayesian approach and using Stan in my GitHub repository https://github.com/ArnauGF/MSc-UPC/tree/bayesian_survival. Moreover, if the reader is interested in an example of a Joint Model fitted using Bayesian software, I would recommend [1]. In the latter reference JAGS is used to carry out several Survival Analysis models.

Despite the fact that Stan provides a big amount of functions, as well as the possibility of build your own functions, working with Stan to fit joint models might be a hard task. Then, we have decided to use the R package *JMbayes2*. This is a great R package developed by Dimitris Rizopoulos et al, from the Erasmus University of Rotterdam. See further information about the package in [12], or in the GitHub repository <https://github.com/drizopoulos/JMbayes2>. *JMbayes2* fits joint models using Markov chain Monte Carlo algorithms implemented in C++. In addition, the package has plenty of functions to summarize and visualize the results of the Bayesian models fitted.

8 Analysis of the AIDS Dataset

The AIDS data set is data from a randomized clinical trial in which both longitudinal and survival data were collected to compare the efficacy and safety of two antiretroviral drugs in treating patients with HIV who had failed or were intolerant of zidovudine (AZT) therapy. The data is available in the *JMbayes2* R package. The fact that this dataset is fully available in a R package makes it a very good choice to put into practice the concepts exposed in this work.

The variables of the AIDS data set are:

- **patient**: patients identifier; in total there are 467 patients.
- **Time**: the time to death or censoring.
- **death**: a numeric vector with 0 denoting censoring and 1 death.
- **CD4**: the CD4 cells count.
- **obstime**: the time points at which the CD4 cells count was recorded.
- **drug**: a factor with levels ddC (baseline) denoting zalcitabine and ddI denoting didanosine.
- **gender**: a factor with levels female (baseline) and male.
- **prevOI**: a factor with levels AIDS denoting previous opportunistic infection (AIDS diagnosis) at study entry, and noAIDS (baseline) denoting no previous infection.
- **AZT**: a factor with levels intolerance (baseline) and failure denoting AZT intolerance and AZT failure, respectively.

The reader can see some observations of our data set in the appendix, Table 8 and 9. We have a total of 467 patients in the study. There is only one longitudinal covariate in our data, it is the CD4. The CD4 cells are counted during one and a half years period, the time points of the measurements are 0, 2, 6, 12, 18. Hence, the longitudinal sub-model will be built to explain this CD4 variable.

8.1 Exploratory data analysis

In this section we do a brief exploratory data analysis. Let us see if the key variables in our problem are balanced in the dataset. In Table 1 the reader can observe a frequency table of the categorical variables of the AIDS dataset. As can be seen, the drug variable, which is the most important due to the fact that our main goal is to compare the effects of the two treatments ddC (zalcitabine) and ddI (didanosine), is balanced (237 and 230 observations respectively). The sex of the patients is not that balanced, we have much more males than females in the study. It is important to highlight that the majority of times are right-censored (418, 89.51%),

drug	gender	prevOI	AZT	event
ddC: 237	female: 45	noAIDS: 160	intolerance: 292	0: 418
ddI: 230	male: 422	AIDS: 307	failure: 175	1: 49

Table 1: Frequency table for the categorical variables

while only few times are observed (49, 10.49%). This is an important feature for the survival outcomes of the model.

Regarding the Time variable, indicating the time to death or censoring, we have the summary statistics: $min = 0.47$, $mean = 12.63$ and $max = 21.40$. The time, in all the temporal variables, is measured in months. Now, we have to do an exploration for the longitudinal variable, CD4. Let us start by plotting the so-called *spaghetti* plot, see Figure 1. As the reader can notice in the *spaghetti* plot, we have a big amount of missing data that should be studied. In addition, by painting in red those patients with no previous infection, and in blue those patients with previous infection, we can see that it seems to be a difference between these two groups. Among the individuals with no previous infections the CD4 cell count seems to be higher, this suggest using the prevOI variable in the longitudinal sub-model.

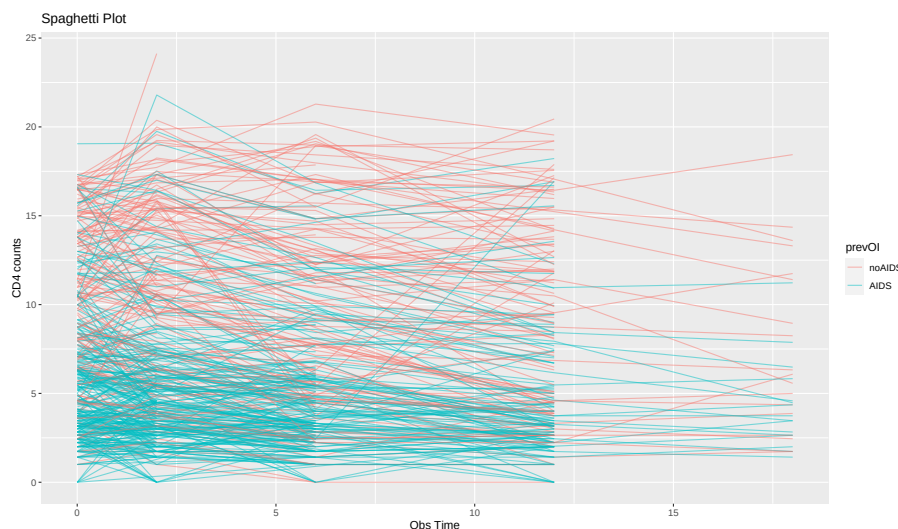


Figure 1: *Spaghetti* plot for the CD4 variable, difference between the prevOI possible outcomes, noAIDS (red) and AIDS (blue), are added.

As said above, we missing data in our longitudinal variable. This occurs usually in studies such the one we are dealing with, in which measurements are done during a period of time. Patients often miss some of their visits during the study. Let us study in depth the number of missings per visit, in Table 2 we can see the number of missing CD4 counts per month. As we can see the missings increase per month, which can suggest that a drop out scenario could be suitable. In addition, other summary statistics for the CD4 variable are provided in Table 2.

obstime	Variable	n	Min	Max	Median	Q1	Q3	mean	Missing data
0	CD4	467	0.00	19.23	6.08	3.32	10.44	7.13	0 (0%)
2	CD4	368	0.00	24.12	5.61	3.32	10.60	7.33	99 (21.2%)
6	CD4	310	0.00	21.28	5.10	2.83	9.38	6.59	157 (33.6%)
12	CD4	226	0.00	20.45	5.00	2.83	10.94	7.03	241 (51.6%)
18	CD4	34	1.41	18.44	4.53	2.69	8.15	6.13	433 (92.7%)

Table 2: Summary statistics for the CD4 variable for each month.

8.2 Longitudinal sub-model

The first step is to choose a longitudinal model to explain the time dependent outcome CD4. Looking at Figure 1, it is clear that the prev0I is explaining part of the variability between subjects. Thus, makes sense to add this variable as a fixed effect in our longitudinal sub-model. Moreover, a random slope seems to be suitable. Then, we have fitted the following longitudinal sub-model

$$y_i(t) = (\beta_0 + b_{i0}) + (\beta_1 + b_{1i})t + \beta_2 \text{prev0I}_i + \epsilon_i(t), \quad (18)$$

where $\epsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$ and $\mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D})$. In (18), t is playing the temporal role and is referring to the obstime variable.

8.3 First Joint Model

Having selected the longitudinal sub-model, it is time to build the survival sub-model and then, the corresponding Joint Model. We start by fitting the most simple Joint Model possible, it is the one adding directly $m_i(t)$ with the association parameter α . We use this model to select the variables that will be used from now on in the different joint models. The covariates finally selected are drug, AZT and gender. The model is the following one:

$$h_i(t) = h_0(t) \exp \{ \gamma_1 \text{drug}_i + \gamma_2 \text{AZT}_i + \gamma_3 \text{gender}_i + \alpha m_i(t) \}, \quad (19)$$

where the baseline hazard is approximated by means of splines, as commented before. The deviance information criterion (DIC) for this model is 8580.404 for the marginal model, and 10297.219 for the conditional model. Besides the DIC, widely applicable information criterion (WAIC) has been computed and we have obtained 8559.045 and 10028.361 respectively. Other models have been fitted using other covariates, but all of them have bigger, and then worse, DIC and WAIC. Hence, we have selected the model in (19). So, from now on, we use the covariates added in (19) and (18).

The model in (19) has been fitted using the *JMbayes2* package. Cubic B-splines has been used. It is important to report the parameters used regarding the Markov Chain Monte Carlo (MCMC) Methods: four chains have been used, as well as 1000 warm up iterations, 2 as a thinning of the chain parameter and a total amount of 6000 iterations. Obtaining a total amount of 10000 simulations for the sample.

A summary of the model (19) is provided in Table 3. As the reader can see, we have Rhat's no much larger than 1 for all the parameters, suggesting that the chains have been successfully mixed and we can use the sample. Trace plots to check how the Markov Chains are mixing are added in the Appendix, see Figure 5. Trace plots suggest that the chains are mixing well.

	Mean	Std Deviation	2.5%	97.5%	p	Rhat
γ_1	0.3428	0.2026	-0.0542	0.7271	0.0900	1.0044
γ_2	0.2992	0.1624	-0.0227	0.6160	0.0704	1.0027
γ_3	-0.3480	0.2606	-0.8452	0.1898	0.1874	1.0052
α	-0.2766	0.0360	-0.3494	-0.2090	0	1.0016
β_0	10.4029	0.3337	9.7434	11.0609	0	1.0013
β_1	-0.1799	0.0158	-0.2111	-0.1498	0	1.0029
β_2	-4.8893	0.4023	-5.6833	-4.0986	0	1.0002
σ	1.7490	0.0473	1.6568	1.8432	0	1.0038

Table 3: Summary statics for the joint model (19).

Summary statistics for all the parameters involved in the model are also exposed in Table 3. Observe that for the treatment variable (γ_1) we have a positive mean estimate, meaning that $\exp(\gamma_1) > 1$, and then for those patients receiving the ddI (didanosine) treatment the instantaneous risk of failure is $\exp(\text{mean}(\gamma_1)) = \exp(0.3428) = 1.4$ times larger than the risk for those patients with ddC (zalcitabine), where all the other covariates are equal. Moreover, the estimation for the association parameter α is highly significant and show a negative association between the survival and the longitudinal outcome. We can plot the posterior distribution of the hazard ratio (HR) for the drug variable ($\exp(\gamma_1)$), in addition to the posterior distribution for the association parameter α . The reader can observe in Figure 2 that the 80% credibility interval for the HR of drug does not include the 1, meaning that there is a significant increase in the instantaneous risk for those patients under the ddI treatment with respect the individuals under the ddC treatment.

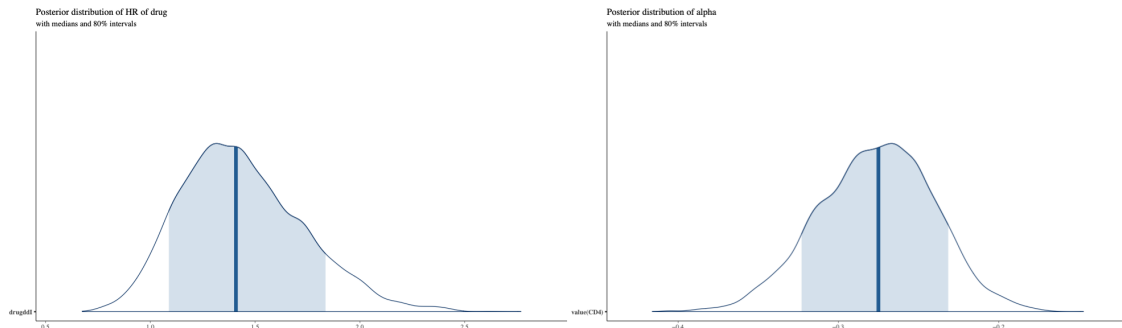


Figure 2: Posterior distributions of the HR of γ_1 (it is, $\exp(\gamma_1)$), and for the association parameter α .

Finally, let us compute the dynamic individualized predictions for this model. We have computed dynamic predictions for the patient number 26. The different predictions for the

longitudinal outcomes are exposed in Figure 3. As the longitudinal information for the 26-th subject is updated, predictions are more accurate and the blue band width decreases.

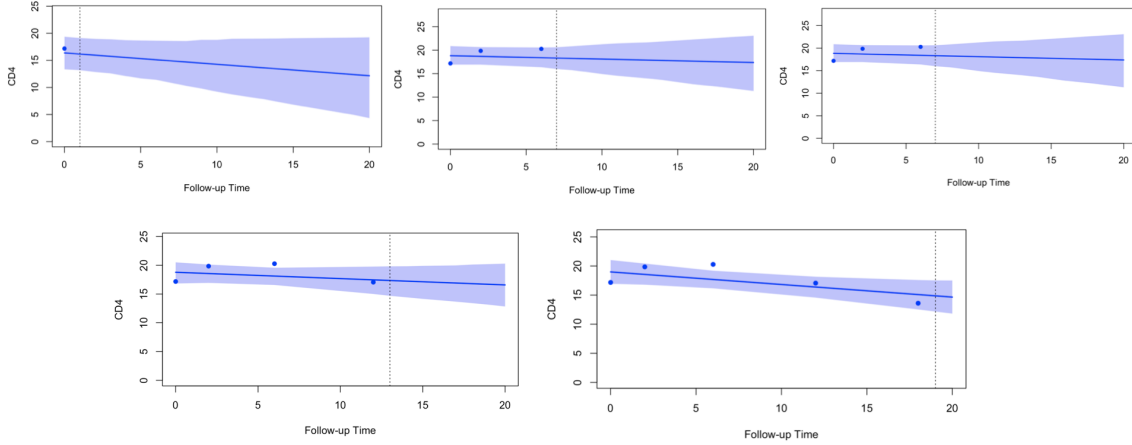


Figure 3: Sequence of different dynamical predictions on the longitudinal outcomes for the subject 26.

Predictions for the survival outcome have been also computed for the patient number 26. The reader can see a similar sequence than the one above in the appendix in Figure 6. In these graphics the predictions for the longitudinal outcomes are plotted jointly with the predictions for the cumulative risk, in such a manner that the evolution of both can be easily analyzed.

8.4 Playing with the functional forms

In this section several joint models with different functional forms will be fitted. As commented before, the covariates to be used are fixed, and they will be the same in all the following models. Then, the models will be compared by means of suitable Bayesian metrics. The best model with respect these quantities will be chosen, and an analysis in depth of this model will be provided. However, in all the models the posterior distribution of the hazard ratio associated with the drug variable will be studied (remember that our main goal is to assess which treatment is preferable).

- Time-dependent slopes model:

We start fitting the value plus slope model (see (12)). Taking into account the variables we are using, the model is as follows

$$\begin{aligned}
 y_i(t) &= m_i(t) + \epsilon_i(t) = (\beta_0 + b_{i0}) + (\beta_1 + b_{1i})t + \beta_2 \text{prevOI}_i + \epsilon_i(t) \\
 \epsilon_i(t) &\sim \mathcal{N}(0, \sigma^2) \quad \text{and} \quad \mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}) \\
 h_i(t) &= h_0(t) \exp \{ \gamma_1 \text{drug}_i + \gamma_2 \text{AZT}_i + \gamma_3 \text{gender}_i + \alpha_1 m_i(t) + \alpha_2 m'_i(t) \},
 \end{aligned} \tag{20}$$

where with respect (19) only changes the functional form used. In this case we have changed the control parameters to achieve the convergence in a properly manner. We have

used only one degree splines, because this lowers the number of parameters by quite a lot, and help us to achieve the convergence of the MCMC methods. In addition, four chains, 2000 warm-up iterations, 4 thinning, and a 12000 iterations per chain have been used. See a summary of the results Table 4. Looking at the Rhat column the reader can see that convergence has been reached. In addition, trace plots are provided in Figure 7, in these graphics we can see that the chains for the association parameters α_1 and α_2 are not that well mixed. In fact, these parameters were the problematic ones when trying to reach the convergence. Nonetheless taking into account the Rhat results, and that the chains are not so badly mixed, we use the sample provided by the MCMC.

	Mean	Std Deviation	2.5%	97.5%	p	Rhat
γ_1	0.4242	0.2516	-0.0696	0.9256	0.0878	1.0037
γ_2	0.3331	0.2358	-0.1145	0.8096	0.1398	1.0126
γ_3	-0.6033	0.3931	-1.4209	0.1422	0.1110	1.0083
α_1	-0.3866	0.0653	-0.5349	-0.2752	0	1.0271
α_2	-5.3678	1.8920	-9.4206	-1.9447	0	1.0721
β_0	10.4524	0.3315	9.7964	11.0995	0	1.0005
β_1	-0.2174	0.0180	-0.2523	-0.1810	0	1.0194
β_2	-4.9159	0.4056	-5.7195	-4.1159	0	1.0011
σ	1.7532	0.0477	1.6626	1.8499	0	1.0041

Table 4: Summary statics for the time-dependent slopes model.

Regarding the summary statistics about the posterior distributions of the parameters, we can see that the γ_1 posterior distribution has $1 - p = 1 - 0.0878 = 0.9122$ of probability above 0. So we can conclude that γ_1 is significantly positive, which implies that the risk of death is greater for those patients under ddI treatment. Hence, this result supports the one obtained with the joint model (19).

- Cumulative effects model:

We now fit the cumulative effects model (see (13)). The model is the following

$$\begin{aligned}
 y_i(t) &= m_i(t) + \epsilon_i(t) = (\beta_0 + b_{i0}) + (\beta_1 + b_{1i})t + \beta_2 \text{prev0I}_i + \epsilon_i(t) \\
 \epsilon_i(t) &\sim \mathcal{N}(0, \sigma^2) \quad \text{and} \quad \mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}) \\
 h_i(t) &= h_0(t) \exp \left\{ \gamma_1 \text{drug}_i + \gamma_2 \text{AZT}_i + \gamma_3 \text{gender}_i + \alpha \int_0^t m_i(s) ds \right\},
 \end{aligned} \tag{21}$$

The control parameters chosen in this case are: four chains, 1000 warm-up iterations, two number of thinning, and a total amount of 10000 iterations per chain. In Table 5 a summary of the results is exposed. By looking at the Rhat values we can see that the Markov Chains are well mixed. Moreover, trace plots are provided in the Appendix in Figure 8.

Looking at the results we can see that the γ_1 posterior distribution has $1 - p = 1 - 0.0942 = 0.9058$ of probability above 0. Thus, we can conclude that γ_1 is significantly greater than 0,

	Mean	Std Deviation	2.5%	97.5%	p	Rhat
γ_1	0.3353	0.1999	-0.0521	0.7204	0.0942	1.0014
γ_2	0.3056	0.1595	-0.0074	0.6204	0.0544	1.0012
γ_3	-0.3259	0.2514	-0.7979	0.1914	0.1970	1.0003
α	-0.2525	0.0323	-0.3193	-0.1928	0	1.0073
β_0	10.3875	0.3299	9.7415	11.0436	0	0.9999
β_1	-0.1699	0.0154	-0.2000	-0.1398	0	1.0022
β_2	-4.8774	0.4018	-5.6672	-4.0954	0	1.0003
σ	1.7600	0.0486	1.6679	1.8574	0	1.0128

Table 5: Summary statics for the cumulative effects model.

so in terms of hazard ratio it is significantly greater than 1, meaning that for those patients under the ddI treatment the instantaneous risk of death is greater than the risk for those individuals receiving the baseline treatment (ddC).

- Value and area model.

Let us fit a model taking into account the present value of the longitudinal process plus the history of the process by means of the integral. It is, a mix between the simple joint model and the cumulative effects model. The model is the following

$$\begin{aligned}
 y_i(t) &= m_i(t) + \epsilon_i(t) = (\beta_0 + b_{i0}) + (\beta_1 + b_{1i})t + \beta_2 \text{prev0I}_i + \epsilon_i(t) \\
 \epsilon_i(t) &\sim \mathcal{N}(0, \sigma^2) \quad \text{and} \quad \mathbf{b}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}) \\
 h_i(t) &= h_0(t) \exp \left\{ \gamma_1 \text{drug}_i + \gamma_2 \text{AZT}_i + \gamma_3 \text{gender}_i + \alpha_1 m_i(t) + \alpha_2 \int_0^t m_i(s) ds \right\}.
 \end{aligned} \tag{22}$$

In order to fit this model and reach the convergence of the MCMC methods, we have used a degree two for the B-splines, four Markov Chains, 2000 burn-in (or warm-up) iterations, two as a number of thinning, and a total amount of 15000 iterations per chain. Using this parameters we have obtained the results in Table 6, as the reader can see in the Rhat column we have obtained good results in terms of convergence. However, the value for α_1 is maybe the only one questioning the convergence of the sampling. Nonetheless, after several attempts this is the best result we have obtained and we have decided to carry on with it. The reader can see the correspondent trace plots to several parameters in the appendix on Figure 9. We can see that the worst mixed chains are the ones corresponding to the association parameters α_1, α_2 , as we expect due to the Rhat values. Notwithstanding, the chains are mixed enough to use the sample.

In Table 6 the summary statistics of the posterior distributions are presented. We can see that the parameter associated to the treatment covariate (drug) is again positive, meaning that there is a significant difference between the instantaneous risk due the treatments. Then, in all the fitted models we have drawn the same conclusion with respect the effect of the treatments, suggesting that the difference is actually a reality.

	Mean	Std Deviation	2.5%	97.5%	p	Rhat
γ_1	0.3571	0.2039	-0.0422	0.7524	0.0793	1.0012
γ_2	0.2938	0.1729	-0.0511	0.6303	0.0935	1.0044
γ_3	-0.3804	0.2681	-0.8910	0.1631	0.1654	1.0010
α_1	-0.5658	0.1889	-0.9289	-0.2221	0.0085	1.0622
α_2	0.2739	0.1767	-0.0500	0.6225	0.1001	1.0538
β_0	10.4228	0.3326	9.7710	11.0788	0	1.0002
β_1	-0.1892	0.0167	-0.2220	-0.1566	0	1.0165
β_2	-4.9026	0.4078	-5.7043	-4.0940	0	1.0006
σ	1.7508	0.0487	1.6594	1.8487	0	1.0129

Table 6: Summary statics for the slope and area model.

Finally, we compare all the models fitted and we choose the best one. As can be observed in the different tables reporting the summary statistics for the posteriors in all the models fitted, the results obtained are similar. This was to be expected taking into account that we have only changed the functional form in the different models. Thus, similar conclusions regarding the inference on the parameters can be drawn from the different models. Nonetheless, it is interesting to assess which is the best model according to several Bayesian information criteria. Deviance information criterion (DIC), widely applicable information criterion (WAIC) and the logarithm of the pseudo marginal likelihood (LPML) are used. In Table 7, these quantities for each model fitted are exposed. In the case of DIC and WAIC, the smaller the better, then the time-dependent slopes model is the best one with respect these quantities. In addition, LPML is the greater the better, thus, the time-dependent slopes model is again the best one. In conclusion, the time-dependent slopes model (20) is the best one, and will be the one used to do some inference.

Model	DIC	WAIC	LPML
Simple (19)	10295.419	10025.448	-5392.622
Time-dependent slopes (20)	10146.831	9894.384	-5358.099
Cumulative effects (21)	10285.139	10021.015	-5445.237
Value and area (22)	10269.618	10004.686	-5408.746

Table 7: Several information criteria for the different joint models fitted.

8.5 Inference with the best model

Finally, in this last section we do some inference with the best model according to the Bayesian information criterions, which is the time-dependent slopes model (20). The summary statistics of the posterior distributions of this model is exposed above in Table 4. The first thing to analyze is the γ_1 parameter, which is the one associated to the drug covariate. In the top left part of Figure 4, we can observe the posterior distribution of the hazard ratio of γ_1 . As we can see the credibility 80% interval is above 1, meaning that there is an increase of

the instantaneous risk of death for those patients under the treatment ddI (didanosine) with respect those patients receiving the ddC (zalcitabine) treatment. Thus, the ddC treatment seems to be better than the ddI.

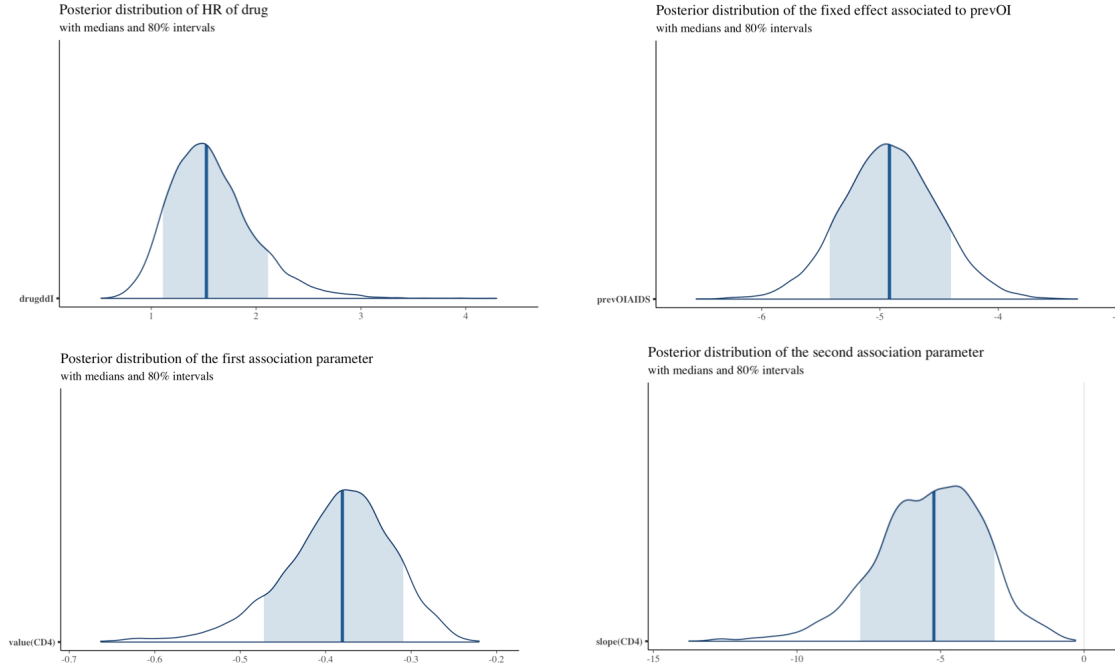


Figure 4: In the top left: posterior distribution of the hazard ratio of γ_1 . Top right: posterior distribution of the fixed effect β_2 . Bottom: in the left the posterior distribution of the association parameter α_1 and in the right the posterior distribution of the association parameter α_2 .

Other posterior distributions of interest are provided in Figure 4. In the top right part of the latter figure we can observe the posterior distribution of the fixed effect β_2 , it is, the fixed effect in the longitudinal sub-model associated with the `prevOI` variable. As we can see, the 80% credibility interval is, approximately, $(-5.5, -4.5)$. This indicates that the posterior distribution of β_2 is taking big negative values. The interpretation of this fact is the following: for those patients having previous AIDS diagnosis the CD4 counts is smaller than for those patients with no previous infection. Notice that this conclusion is supporting what we saw in Figure 1 during the exploratory analysis.

Moreover, in the bottom part we have the posterior distributions of the association parameters α_1, α_2 . Both show negative association between the event and the longitudinal outcome. Meaning that a decrease on the longitudinal outcome implies an increase of risk of death. Furthermore, the association parameter α_2 associated with the slope, is the one showing an stronger association, since is more negative than the other parameter α_1 . In addition, taking into account the mean of both posteriors distributions (see Table 4), we can see that a decrease of one unit in the CD4 cells correspond to an increase of $\exp(-\text{mean}(\alpha_1)) = 1.472$ times in the instantaneous risk of death (i.e. a 47.2% increase).

In addition, we have checked which are the patients with bigger mean random effects in absolute value. These patients are the more distant with respect the "mean patient", they

are the patients 189 for the random intercept b_{i0} and 286 for the random slope b_{i1} , and the values are 14.83 and 0.735, respectively. Other quantities such as the dynamical individualized predictions can be computed as we have done in previous sections. Nonetheless, due the extension of the work these quantities and other are not added.

9 Conclusions and future research

Joint models have proven to be a very useful tool when working with time to event and longitudinal data simultaneously. The mix of these different type of data produces complex scenarios that should be faced with advanced methods. Regarding the theoretical part of the work, despite the fact that joint models are not straightforward objects, the theoretical framework behind it is not that difficult to understand, and it has been exposed during the work in a clear manner. I would like to put forward that the main motivation of this work was a theoretical motivation. It is, to understand the mathematics behind joint models, and study how Bayesian analysis is used when working with advanced topics of survival analysis. I believe that these theoretical objectives have been met.

Moreover, in practice, joint models have helped us to demonstrate the efficacy of one treatment over another. In all the models fitted in the case study part, we have drawn the same conclusion about the better effect of the ddC treatment compared with the ddI treatment. Then, our study has been useful to statistically demonstrate than one treatment is better than the other. In addition, doing this taking into account the effect of the longitudinal outcome would not have been possible without the use of joint models. Other features of joint models under the Bayesian approach, such as dynamic individualized predictions or the interpretation of the association parameters, help us gaining a good understanding on the problem.

This work leaves me with a number of issues to address in the future. First and foremost, I would like to use in practice Bayesian model averaging techniques, and assess how these techniques can be used not only for prediction, but for inference. In this work I have not been able to do it, because to do it I needed to use the likelihood of the model (10), and as it is so extensive and with so many parameters it has given me some technical problems that I have not been able to solve. In addition, how to compute the marginal likelihoods of the models (16) by simulation is another interesting topic that I would like to deal with in the future. Moreover, taking into account the length of the work without including this part in practice, I preferred to leave it for future research. So, I hope to be able to deal with this problem or something similar soon.

This final assignment of the course has been a good opportunity to learn about a topic of my interest, and I am proud about the results obtained. My knowledge on Bayesian analysis applied to survival analysis, as well as longitudinal analysis has incredibly grown after carry out this project. This is a topic that I would like to investigate further in a future, so it has been great to be able to discuss this topic for the final project.

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A Appendix

	patient	Time	death	CD4	obstime	drug	gender
1	1	16.97	0	10.68	0	ddC	male
2	1	16.97	0	8.43	6	ddC	male
3	1	16.97	0	9.43	12	ddC	male
4	2	19.00	0	6.32	0	ddI	male
5	2	19.00	0	8.12	6	ddI	male
6	2	19.00	0	4.58	12	ddI	male

Table 8: First six observations of the AIDS dataset

prevOI	AZT	start	stop	event
AIDS	intolerance	0	6.00	0.00
AIDS	intolerance	6	12.00	0.00
AIDS	intolerance	12	16.97	0.00
noAIDS	intolerance	0	6.00	0.00
noAIDS	intolerance	6	12.00	0.00
noAIDS	intolerance	12	18.00	0.00

Table 9: First six observations of the AIDS dataset

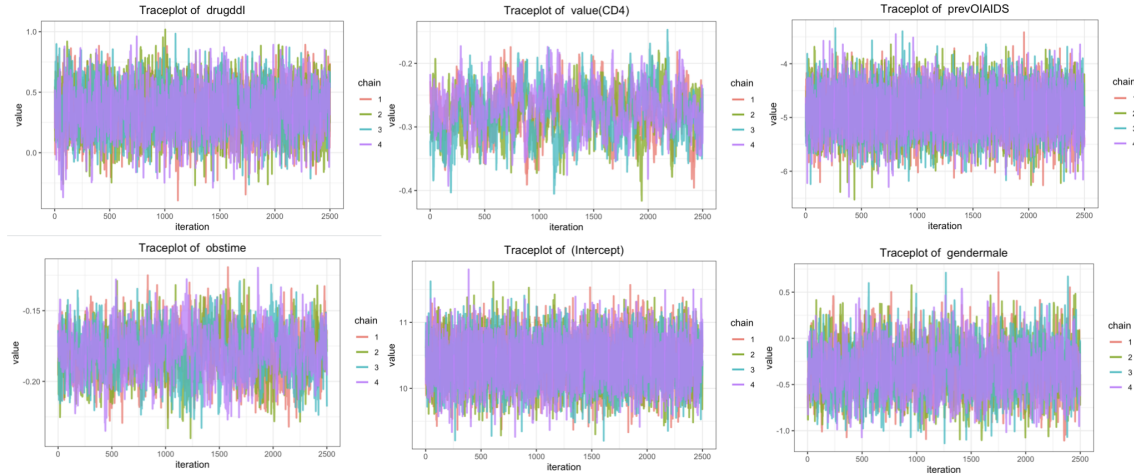


Figure 5: Trace plots of (19). Only the iterations after warm-up are plotted.

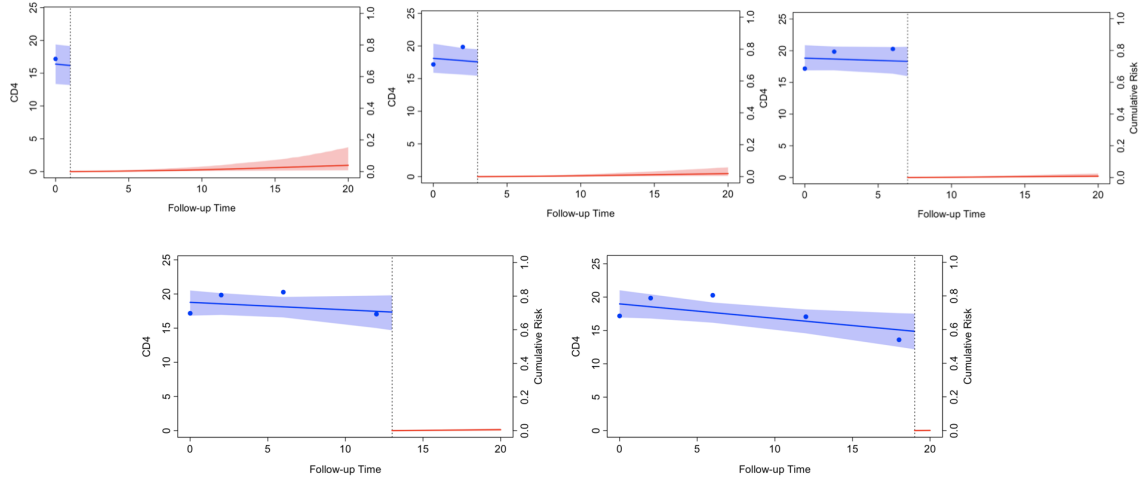


Figure 6: Sequence of different dynamical predictions on the longitudinal and survival outcomes for the subject 26.

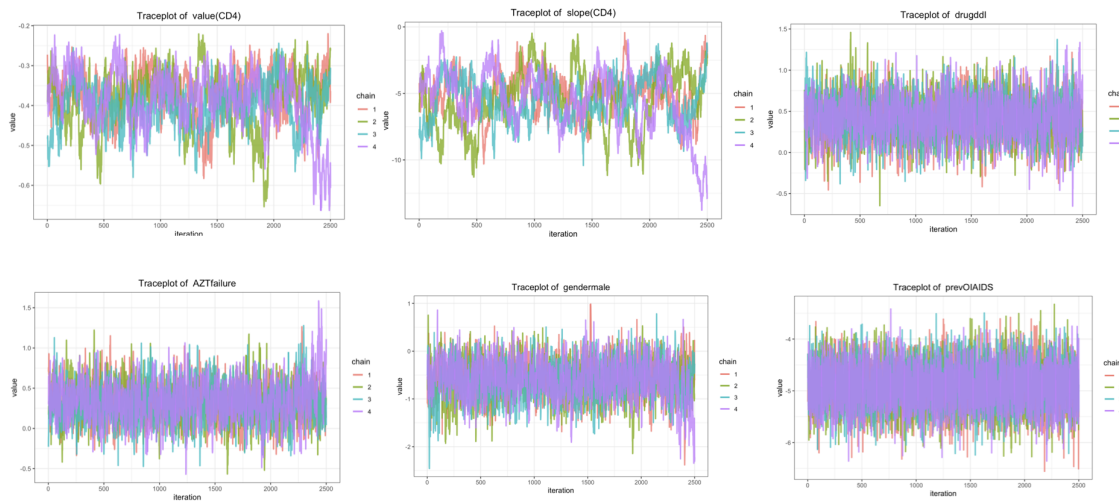


Figure 7: Trace plots for the model (20). Only the iterations after warm-up are plotted (2500 iterations per chain are plotted).

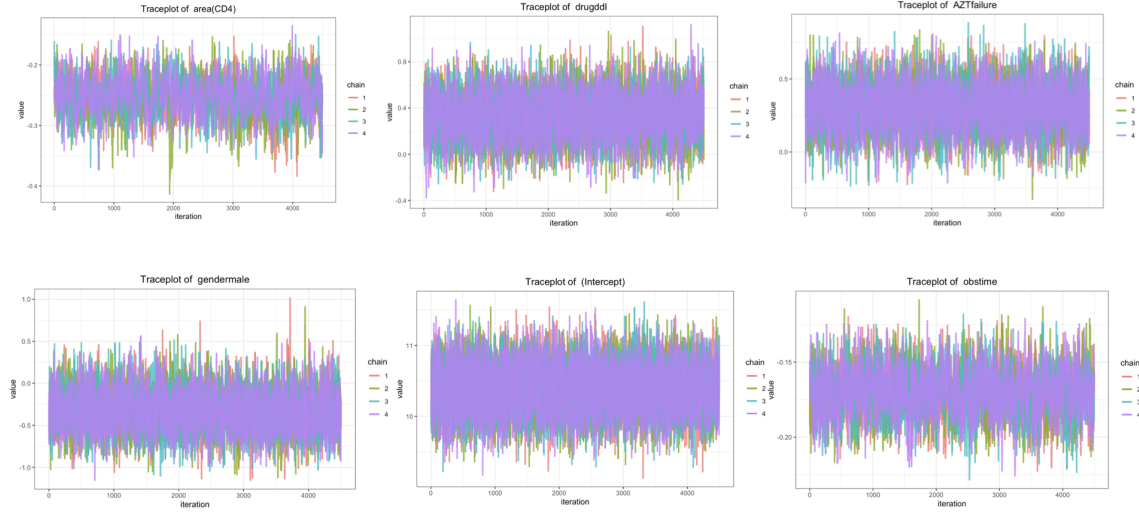


Figure 8: Trace plots for the model (21). Only the iterations after warm-up are plotted.

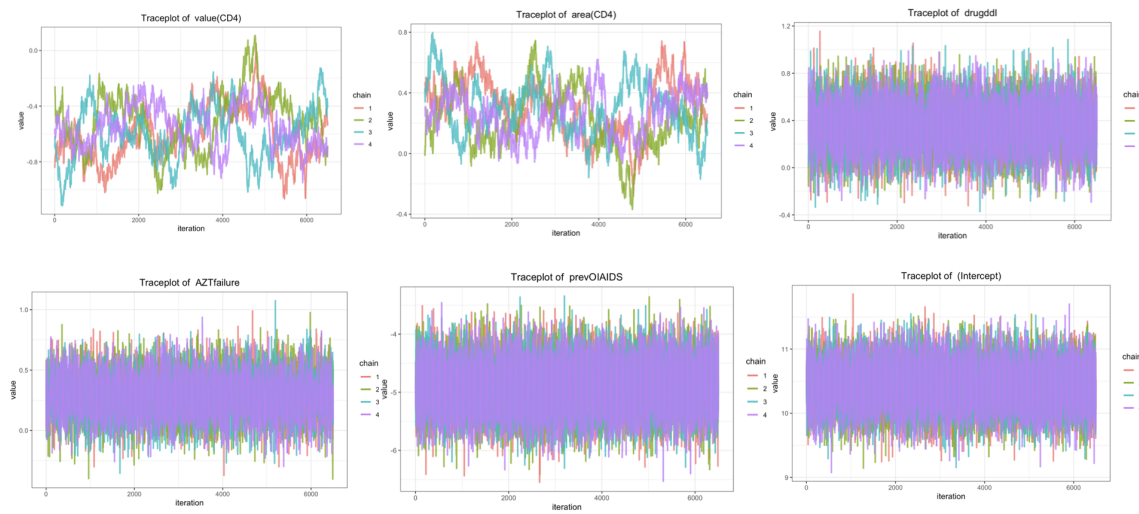


Figure 9: Trace plots for the model (22). Only the iterations after warm-up are plotted.